Operating Instruction

for

Float-Type Flow Meter

and Switch

Model: DSV-...

1. Contents

Manufactured and sold by:

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2. Note

Please read these operating instructions before unpacking and putting the unit into operation. Follow the instructions precisely as described herein.
The devices are only to be used, maintained and serviced by persons familiar with these operating instructions and in accordance with local regulations applying to Health & Safety and prevention of accidents.

When used in machines, the measuring unit should be used only when the machines fulfil the EWG-machine guidelines.

as per PED 97/23/EG
In acc. with Article 3 Paragraph (3), "Sound Engineering Practice", of the PED 97/23/EC no CE mark.
Table 8, Pipe, Group 1 dangerous fluids

3. Instrument Inspection

Instruments are inspected before shipping and sent out in perfect condition. Should damage to a device be visible, we recommend a thorough inspection of the delivery packaging. In case of damage, please inform your parcel service / forwarding agent immediately, since they are responsible for damages during transit.

Scope of delivery:
The standard delivery includes:

- Float-Type Flow Meter / Switch model: DSV
- Operating Instructions

4. Regulation Use

The model DSV is employed for the measurement and monitoring of liquid flows which are homogeneous, free from impurities and have a low viscosity. In addition, they must be compatible with the materials used in fabrication of these units. High viscosity media can introduce considerable measurement errors. Also large dust particles can block the float, resulting in error-messages and erroneous measurements. In addition, ferrite particles, that may deposit on the embedded magnets of the float, can trigger the same effect. To overcome this problem, the installation of a magnetic filter is recommended.

The units are provided as follows:

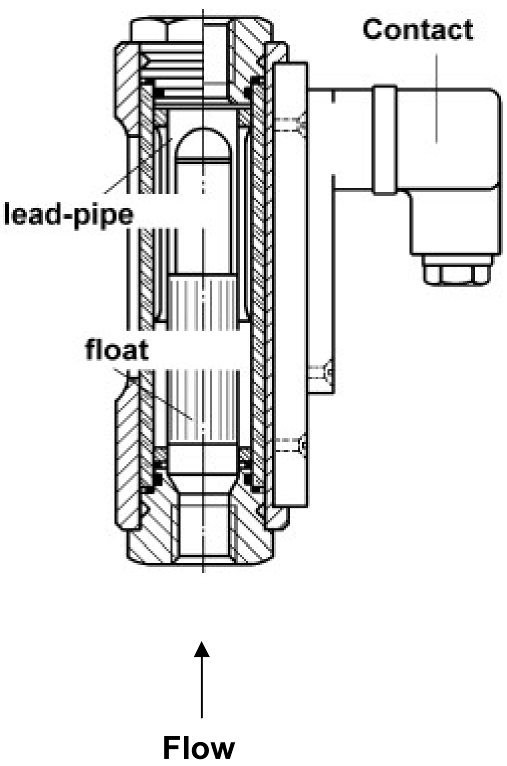
Flow measurement
The reading of the present media flow takes place on site. The top edge of the float indicates the flow directly in l/min (water) on the scale attached to the graduated measuring glass**.**

Limit-contacts (only with model DSV-2... and DSV-3...)
To monitor the flow values, the units are provided with one or two adjustable limit-contact(s).

Standard version:	Normally Open function (with increasing flow)
Special version:	Change-over contact function

The switch contact is adjustable over most of the measuring range of the device, under consideration of the switch hysteresis.

5. Operating Principles



The proven KOBOLD flow meters and switches model DSV are based on the principle of the well-known float-type flow meters except for the conventional tapering measuring tube.
These patented instruments are provided instead with a cylindrical flow tube with conical slots around the periphery. This eliminates the usual problems of guiding the cylindrical float within a tapering measuring glass. The novel design including the provision of an appropriately dimensioned annular gap of constant width between the float and the flow tube has enabled the sensitivity to dirt to be considerably reduced.
The float contains a permanent magnet which actuates a bistable reed contact external to the flow circuit, that is, the flowing medium is hermetically separated from the electrical contact. In addition it is embedded in a height-adjustable switch housing thus ensuring that the contact is protected even in an aggressive atmosphere.
As the medium enters the instrument the float rises. Once its magnetic field reaches the contact tips of the reed switch the contact closes. As the flow increases the float rises further until it reaches its stop. This prevents the float from going beyond the contact range of the magnetic operating tube, that is, the contact remains closed thus ensuring bistable switching. The top edge of the float serves to indicate the flow on the measuring glass in L/min.

6. Mechanical Connection

Before installation:

- Make sure that the permitted max. operating pressure and temperature limits are not exceeded (see section 10. Technical Information)
- Remove all transport safety locks and ensure that no remains of packing material reside inside the unit.
- The sealing of union fittings is carried out through Teflon tape or similar material.
- During installation of these devices, attention must be paid that no large course or pressure-load is exercised on the union fittings. We recommend fastening the inlet and outlet line at approx. 50 mm distance mechanically, from the union fittings.
- The units may not be installed at a location within an inductive field.

Attention! The union fittings of these devices must be held fixed by means of a suitable wrench while screwing in. Otherwise, stresses will transfer to the housing which may lead to the destruction of graduated measuring glass.

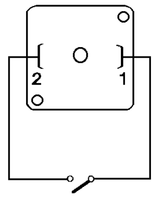
- If possible, it should be checked after mechanical installation whether the connections between union fittings and pipes are properly sealed

Attention! While installing in a free and open environment, please pay attention to the fact that the freezing of medium can destroy the measuring tube.

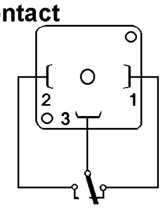
7. Electrical Connection

7.1. DSV-2... and DSV-3

N/O Contact



change over contact



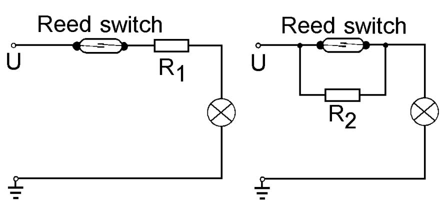
- Make sure that the electrical supply wires are powerless.
- Undo the locking screw on the plug cap and remove the plug from the socket.
- Mount supply line in the plug cap as shown in the wiring diagram.
- If the contact has not been adjusted, adjust it now. (see section 8. Commissioning).
- Place the plug on the socket and fasten it with the safety-screws. (see section 8. Commissioning).

Attention! Every single specified electrical value for the sealed contact should not be exceeded even for short periods. We recommend contact protection relays or other contact protection device for higher switching values.

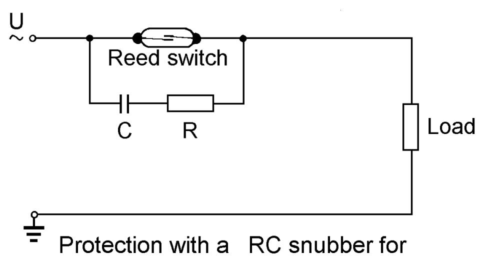
- When the external devices have been connected to the limit contact and the switch point has been set, the electrical connection is complete.
- The instrument can now be placed into operation.

Examples of contact protection devices

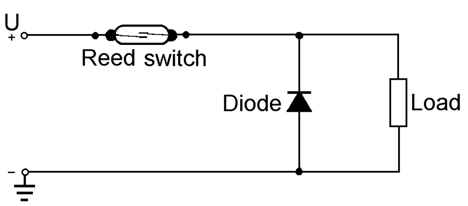
For capacitive and inductive loads, (long leads and relays/contactors) we recommend contact protection relays or the following suppressor circuits.



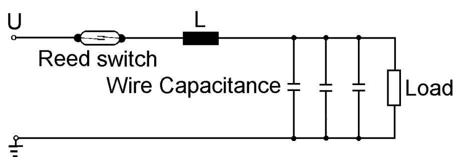
Lamp load with parallel or series resistance to the switch.



Protection with a RC snubber for a.c. current and inductive load



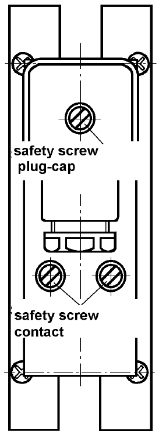
Protection with an idle diode for d.c. current and inductive load.



Protection against high discharge from condensers and load capacitances.

8. Commissioning

Application in machines, according to guidelines 89/392/EWG, the commissioning of these units is barred until it is determined that the machine is in agreement with the designations of guidelines.



Adjustment of limit-values

- Loosen both safety screws on the contact base clown, with the help of a screwdriver.
- Dislocate the contact base down until it reaches the stop.
- Remove the plug-cap after loosening the safety screws.
- Connect with PIN 1 + 2 (change-over contact PIN 2 + 3) with a suitable circuit test unit.

Attention! Sudden opening of the flow supply line can lead to pressure peaks, which repeatedly exceed the operating pressure (water hammer!) This can lead to the breakage of the measuring glass.

Case 1: Device in circuit
Open the supply line.
Let the medium flow slowly, until the top edge of the float shows the flow volume of desired switching point.

Case 2: Device outside circuit
Raise the float with a suitable tool, until the top edge of the float shows the desired flow volume of desired switching point.

- The reed contact is now closed (electrically closed-circuit).
- Move the switch-housing upwards, until the reed-contact is just open (no electrical passage). The contact is now adjusted for decreasing flow. If the contact needs to be adjusted for increasing flow, it must be relocated now around the hysteresis about 3-5 mm downwards.
- Adjust the safety screws in this position by tightening them. Install the plug-cap. The unit is ready for operation.
- On correct adjustment of the limit-contact it has a bistable switching behaviour; i.e. upon exceeding the adjusted limit, the contact remains closed and below the adjusted limit, the contact stays open.

Hysteresis

Hysteresis reflects the difference between switch-on and switch-off points of a contact. Through standard tuning of the float magnet and reed contact power (AW number) a hysteresis of about 3-5 mm float stroke is achieved. Thereby, ensuring that the contact exhibits bistable behaviour.

Exceeding the measuring range

Attention! Continuously exceeding the measuring range, especially with pulsating flow and the resulting contact of the float to the limit-pin, can lead to increase wear and damage to the unit. In such a case, please consult your supplier.

Non-pulsating flow can essentially exceed the measuring range, merely resulting in a notable increase in pressure-loss.
(The maximum operational pressure is not to be exceeded.)

9. Maintenance

In the event that the medium to be measured is free from impurities, the model DSV is almost maintenance free. If lime or other deposits on the measuring glass or within interior parts develop, the units should be cleaned on a regular basis. After removing the upper grub-screw, the screw joint can be withdrawn from the unit, and the inner parts can be extracted for cleaning. The measuring glass can be cleaned with the help of a suitable brush. After cleaning, the unit must be reassembled in the correct sequence.

10. Technical Information

Housing:	Aluminium anodised (not wetted)
Connections:	DSV-x1..: Brass MS 58, nickel-plated DSV-x2..: St. Steel 1.4301
Float:	see section 11. Order Codes
Orifice:	DSV-x1..: Brass MS 58, nickel-plated DSV-x2..: St. Steel 1.3955
Measuring glass:	Duran 50 (Borosilicate glass)
Sealing rings:	DSV-x1..: NBR DSV-x2..: FPM
Max. temperature:	100 °C (Metal-float) 70 °C (PP or PVDF float)
Max. pressure:	10 bar
Accuracy:	± 4% (F.S.)
Mounting position:	vertical, flow from bottom to top

Contacts (DSV-2..., DSV-3...)

Electrical connection: Connector acc. DIN 43 650

Electrical switching values:

N/O contact (CSA)	max. 240 V_{AC} / 100 VA / 1.5 A
Changeover contact (CSA)	max. 240 V_{AC} / 60 VA / 1 A
N/O contact (UL)	250 V_{AC} - 0,4 A / 200 V_{DC} - 0,25 A 50 V_{DC} - 1,0 A
Changeover contact (UL)	250 V_{AC} - 0,136 A / 30 V_{DC} - 1,0 A

Protection: IP 65

11. Order Codes

Example: (DSV-2101H R0 R08)

Flow meter model: DSV-1...

Measuring range L/min. water	Pressure loss Δ P (bar)	Float according to version		Brass	Stainless steel	Contact	Connection	
		Brass	St. steel					
0.25...1.25	0.04	PP	PVDF	**DSV-1101H...**	**DSV-1201H...**	**...00..**= without contacts	**..R08**= G 1/4 **..R15**= G 1/2	**..N08**= 1/4 NPT **..N15**= 1/2 NPT
0.5...2.5	0.06	PP	PVDF	**DSV-1102H...**	**DSV-1202H...**			
1...4.5	0.04	PP	PVDF	**DSV-1103H...**	**DSV-1203H...**			
1...10	0.04	PP	PVDF	**DSV-1104H...**	**DSV-1204H...**		**..R15**= G 1/2 **..R20**= G 3/4	**..N15**= 1/2 NPT **..N20**= 3/4 NPT
2...18	0.07	Brass, nickel-pl.	1.4301	**DSV-1105H...**	**DSV-1205H...**			
2...25	0.08	PP	PVDF	**DSV-1106H...**	**DSV-1206H...**		**..R20**= G 3/4 **..R25**= G 1	**..N20**= 3/4 NPT **..N25**= 1 NPT
2.5...50	0.14	Brass, nickel-pl.	1.4301	**DSV-1107H...**	**DSV-1207H...**			
10...100	0.3	Brass, nickel-pl.	1.4301	**DSV-1108H...**	**DSV-1208H...**		**..R25**= G 1	**..N25**= 1 NPT
10...130	0.4	PP	PVDF	**DSV-1109H...**	**DSV-1209H...**		**..R32**= G 1 1/4	**..N32**= 1 1/4 NPT

Flow meters and switches with 1 contact model: DSV-2...

Measuring range L/min. water	Pressure loss Δ P (bar)	Float according to version		Brass	Stainless steel	Contact	Connection	
		Brass	St. steel					
0.25...1.25	0.04	PP	PVDF	**DSV-2101H...**	**DSV-2201H...**	**...R0...**= 1 N/O contacts **...U0...**= 1 changeover contact **...C0...**= 1 N/O contact (UL) **...D0...**= 1 changeover contact (UL)	**..R08**= G 1/4 **..R15**= G 1/2	**..N08**= 1/4 NPT **..N15**= 1/2 NPT
0.5...2.5	0.06	PP	PVDF	**DSV-2102H...**	**DSV-2202H...**			
1...4.5	0.04	PP	PVDF	**DSV-2103H...**	**DSV-2203H...**			
1...10	0.04	PP	PVDF	**DSV-2104H...**	**DSV-2204H...**		**..R15**= G 1/2 **..R20**= G 3/4	**..N15**= 1/2 NPT **..N20**= 3/4 NPT
2...18	0.07	Brass, nickel-pl.	1.4301	**DSV-2105H...**	**DSV-2205H...**			
2...25	0.08	PP	PVDF	**DSV-2106H...**	**DSV-2206H...**		**..R20**= G 3/4 **..R25**= G 1	**..N20**= 3/4 NPT **..N25**= 1 NPT
2.5...50	0.14	Brass, nickel-pl.	1.4301	**DSV-2107H...**	**DSV-2207H...**			
10...100	0.3	Brass, nickel-pl.	1.4301	**DSV-2108H...**	**DSV-2208H...**		**..R25**= G 1	**..N25**= 1 NPT
10...130	0.4	PP	PVDF	**DSV-2109H...**	**DSV-2209H...**		**..R32**= G 1 1/4	**..N32**= 1 1/4 NPT

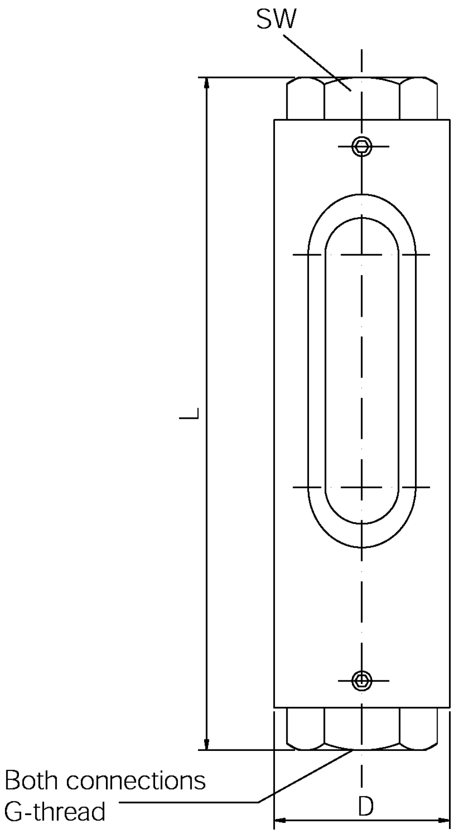
Flow meters and switches with 2 contacts model: DSV-3...

Measuring range L/min. water	Pressure loss Δ P (bar)	Float according to version		Brass	Stainless steel	Contact	Connection	
		Brass	St. steel					
0.25...1.25	0.04	PP	PVDF	**DSV-3101H...**	**DSV-3201H...**	**...RR...**= 2 N/O contacts **...UU...**= 2 changeover contacts **...CC...**= 2 N/O contacts (UL) **...DD...**= 2 changeover contacts (UL)	**..R08**= G 1/4 **..R15**= G 1/2	**..N08**= 1/4 NPT **..N15**= 1/2 NPT
0.5...2.5	0.06	PP	PVDF	**DSV-3102H...**	**DSV-3202H...**			
1...4.5	0.04	PP	PVDF	**DSV-3103H...**	**DSV-3203H...**			
1...10	0.04	PP	PVDF	**DSV-3104H...**	**DSV-3204H...**		**..R15**= G 1/2 ..R20= G 3/4	**..N15**= 1/2 NPT **..N20**= 3/4 NPT
2...18	0.07	Brass, nickel-pl.	1.4301	**DSV-3105H...**	**DSV-3205H...**			
2...25	0.08	PP	PVDF	**DSV-3106H...**	**DSV-3206H...**		**..R20**= G 3/4 **..R25**= G 1	**..N20**= 3/4 NPT **..N25**= 1 NPT
2.5...50	0.14	Brass, nickel-pl.	1.4301	**DSV-3107H...**	**DSV-3207H...**			
10...100	0.3	Brass, nickel-pl.	1.4301	**DSV-3108H...**	**DSV-3208H...**		**..R25**= G 1	**..N25**= 1 NPT
10...130	0.4	PP	PVDF	**DSV-3109H...**	**DSV-3209H...**		**..R32**= G 1 1/4	**..N32**= 1 1/4 NPT

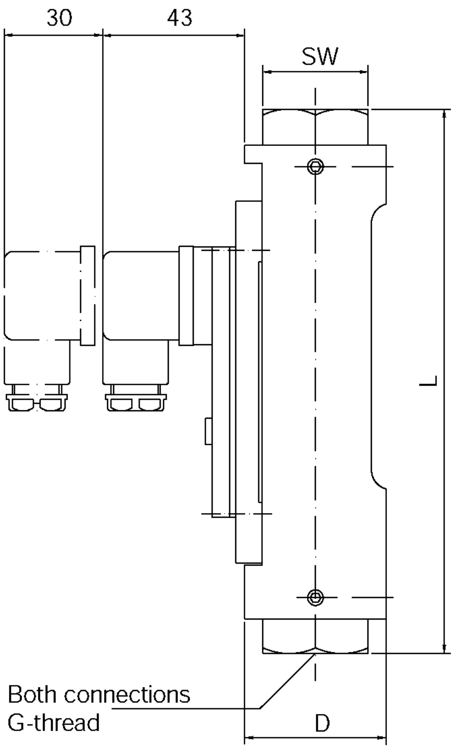
12. Dimensions

Model	SW	G	L (mm)	D (mm)	Weight approx. (kg)			dP bar	Float with unit version	
					DSV-1	DSV-2	DSV-3		brass	st.steel
DSV-..01H	32	1/4 (1/2)	161 (165)	43	0.75	1.0	1.25	0.04	PP	PVDF
DSV-..02H	32	1/4 (1/2)	161 (165)	43	0.75	1.0	1.25	0.06	PP	PVDF
DSV-..03H	32	1/4 (1/2)	161 (165)	43	0.75	1.0	1.25	0.04	PP	PVDF
DSV-..04H	32	1/2 (3/4)	165	43	0.75	1.0	1.25	0.04	PP	PVDF
DSV-.. 05H	32	1/2 (3/4)	165	43	0.75	1.0	1.25	0.07	MS Ni-pl.	1.4301
DSV-..06H	41	3/4 (1)	165 (176)	48	1.0	1.25	1.5	0.08	PP	PVDF
DSV-..O7H	41	3/4 (1)	165 (176)	48	1.0	1.25	1.5	0.14	MS Ni-pl.	1.4301
DSV-..08H	41	1	204	48	1.2	1.45	1.7	0.3	MS Ni-pl.	1.4301
DSV-..09H	46	1 1/4	222	55	1.5	1.75	2.0	0.4	PP	PVDF

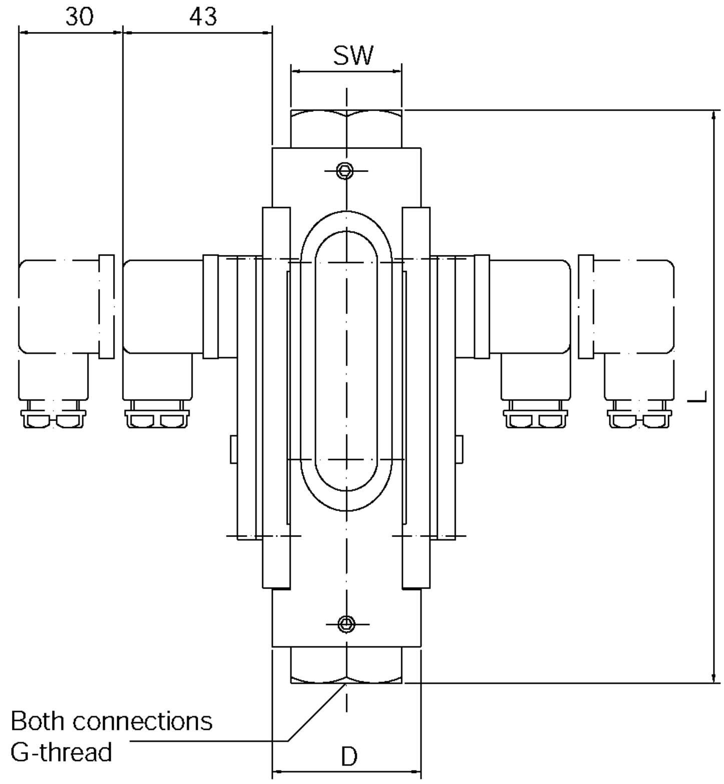
DSV-1...



DSV-2...



DSV-3...



13. Recommended Spare Parts

Only the unit-parts and materials will be listed. Corresponding to the model version, parts in different sizes are available.
(Please ensure you mention the model version, when ordering spare parts).

1.1) Float Brass
1.2) Float Polypropylene
1.3) Float St. Steel
1.4) Float PVDF
2.1) Slotted nozzle Brass
2.2) Slotted nozzle St. Steel
3.1) O-Ring NBR
3.2) O-Ring FPM
4.1) Contact (N/O function) (CSA)
4.2) Contact (changeover function) (CSA)
4.3) N/O (UL)
4.4) Changeover (UL)
5.1) Meas. glass with scale

13 Declaration of Conformance

We, KOBOLD-Messring GmbH, Hofheim-Ts, Germany, declare under our sole responsibility that the product:

Flow Meter / Switch Model: DSV

is in agreement with the following standards:

DIN EN 61010-1 1994-03
Safety requirements for electrical equipment for measurement, control and laboratory use.

DIN VDE 0470-1 1992-11
Protection through housing (IP-code)

Also the following EWG guidelines are fulfilled:
2006/95/EC

97/23/EG PED
Category II, Table 8, pipe, liquids
Group 1 dangerous fluids
Module D, mark CE0098
Notified body: Germanischer Lloyd

Signed:

Date: 18.11.03

ppa. Peters ppa. Wenzel